

# OptiCrop: Solution Architecture

See `opticrop_architecture.png` (in 7.Project Documentation) for the full visual diagram. Summary of components below.

## Components

- Client: browser rendering the HTML form (`index.html`) and result page (`result.html`).
- Server: Flask application (`app.py`) handling routing and inference.
- Model layer: serialized `model.pkl`, `scaler.pkl`, `label_encoder.pkl` loaded at startup.
- Offline pipeline: Jupyter notebook performing EDA, preprocessing, training, evaluation, and clustering, producing the serialized model artifacts.

## Request Lifecycle

- User submits the form -> POST `/predict` -> Flask parses form fields into a `DataFrame` -> `StandardScaler` transforms the input -> model predicts an encoded label -> `LabelEncoder` decodes it to a crop name -> `result.html` renders it.